Nguyen Thai Hung

Da Nang, Viet Nam | hungnguyen.190206@gmail.com | 0353 345 206 | github.com/huanight19RaH
linkedin.com/in/thai-hung-nguyen-000b24356

Introduction

AI undergraduate student at FPT University with a strong interest in computer vision and applied machine learning. Seeking an AI internship to apply programming, research, teamwork, and leadership experience while contributing to practical, innovative products. Proactive learner motivated to grow through real-world engineering and research challenges.

Education

FPT University, Artificial Intelligence – Da Nang, Viet Nam Jan 2024 – present

- Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Data Structures, Probability, Linear Algebra

Experience

AI Developer Intern - IDA Department, FPT Information System (FIS) - R&D – Da Nang, Viet Nam Aug 2026 – present

- Develop real-time computer vision streaming and video ingestion pipelines for the Kamoi Vision AI (AICC) safety and industrial compliance project.
- Implement multi-camera RTSP ingestion, frame quality gating, and sliding GPU queue management to eliminate accumulated processing latency.
- Integrate YOLO detection and ByteTrack/BoT-SORT tracking with Kalman filtering for reliable multi-object tracking and event rule analytics under complex industrial scenes.
- Benchmark model inference performance and collaborate with senior engineers to deploy containerized vision AI microservices.

Achievements and Awards

Code Mosaic 2025 – FPT University Da Nang Oct 2025

Second Prize

- Competed in an annual programming hackathon focused on data structures and algorithms.

FPT Education Research Festival 2026 (ResFes) – FPT University Da Nang May 2026

Consolation Prize

- Awarded in the university-system research festival for high-impact computer vision and deep learning image restoration research.

Research Connect 2026 – FPT University Da Nang May 2026

Consolation Prize

- Recognized in an annual scientific research competition organized by FPT University Da Nang.

Da Nang City Excellent Student Competition in Mathematics – Da Nang, Viet Nam Jan 2024

Consolation Prize

- Awarded in the city-level academic competition organized by the Da Nang Department of Education and Training.

Skills

Programming Languages: Python (intermediate), C/C++ (basic), SQL (basic)

AI / ML Frameworks: PyTorch, TensorFlow, scikit-learn, OpenCV

Data / Tools: pandas, NumPy, Matplotlib, SQL, Git, Docker, Jira, Qdrant, Supabase

AI Areas: Computer Vision, Image Restoration, NLP, Deep Learning, Machine Learning

Languages: Vietnamese, English (IELTS 6.0, 2024)

Projects

Traffic Flow Detection

Video-based traffic analytics system for detecting, tracking, and counting vehicles by lane, class, and direction.

- Contributed ideas to the project workflow and processing pipeline; took primary responsibility for frontend implementation.
- Built user-facing flows for video upload/live-stream analysis, ROI/lane/counting-line configuration, progress tracking, and result visualization.
- Integrated a traffic analysis pipeline using YOLO detection, ByteTrack tracking, lane assignment, direction validation, and line-crossing counting.
- Reported benchmark results on UA-DETRAC held-out data, including Event F1 of **0.942**, WAPE of **5.04%**, batch processing at **75.8 FPS** (**3.03x** real-time), and live streaming at **14.9 FPS** with **0%** drop rate.

Assessing Physics-Informed Priors in Underwater Image Restoration

Accepted conference paper at EIDT (Co-Author) evaluating physics-informed priors for underwater image restoration under a controlled customized U-Net framework.

- Co-authored paper accepted at EIDT conference; conducted initial literature survey, contributed to manuscript writing, revision, and technical formulation.
- Studied transmission map **t(x)** and backscatter map **B(x)** as physics-informed auxiliary channels for underwater image restoration.
- Evaluated RGB, RGB + **t(x)**, RGB + **B(x)**, and RGB + **t(x)** + **B(x)** input configurations under an identical customized U-Net backbone.
- Reported results on UIEB T90 and EUVP-Test, including **21.012 dB PSNR / 0.884 SSIM** on T90 and **26.644 dB PSNR / 0.873 SSIM** on EUVP-Test for the 5-channel model.

HybridLLIENet: Low-Light Image Enhancement

Published conference paper in Springer MIWAI 2026 (Co-Author) on low-light image enhancement using multi-space luminance prior fusion and a U-Net-style restoration network.

- Co-authored paper published in Springer MIWAI 2026; conducted literature survey, contributed to manuscript writing and revision, and supported experimental discussion.
- Contributed to benchmarking low-light enhancement models on LOL datasets using PSNR, SSIM, and LPIPS metrics.
- Studied HybridLLIENet, which fuses LAB-L, HSV-V, and YCbCr-Y luminance priors enhanced by CLAHE and gamma correction with RGB input.
- Reported LOLv2-Real results of **21.23 dB PSNR**, **0.858 SSIM**, and **0.127 LPIPS** the model uses **8.04M** parameters and **79.26 GFLOPs**.

SchoolSummar

Structure-Aware RAG platform for complex scientific documents with grounded citation verification and low-latency retrieval.

- Conducted comprehensive benchmark evaluations of document parsing and OCR engines (Docling, PaddleOCR, PaddleOCR-VL, Surya OCR) on scientific papers from DocLayNet.
- Evaluated CER, WER, Character F1, normalized exact match, page latency, and throughput (chars/sec) to select and optimize the primary OCR pipeline.
- Supported RAG pipeline development, integrating Docling AST hierarchy chunking and spatial bounding box citation receipts `[ymin, xmin, ymax, xmax]` for PDF source verification.
- Contributed to containerized microservices infrastructure and testing using Qdrant Cloud, Supabase PostgreSQL, and FastAPI/ Next.js.

Certificates

- AI Foundations for Everyone Specialization - IBM, Coursera
- AI For Everyone - DeepLearning.AI, Coursera
- Ethics in the Age of AI - LearnQuest, Coursera

Activities

FU-Dever Club – FPT University Da Nang Jan 2026 – Oct 2026

Former Vice President

- Helped lead an academic technology club for students interested in information technology, software engineering, and AI.
- Organized learning events, technical workshops, and coding activities for members and students outside the club.

Bat Nu Cuoi Len Charity Project - Sunshine Team – Da Nang, Viet Nam Mar 2026

Media Lead

- Led media activities for a charity project supporting Village of Hope in Da Nang.
- Created and published social media posts, managed the Sunshine Team fanpage, and participated in the charity event.